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A novel one-shot forming technology for ultra-high strength multi-material structures: Process design, simulation and proof of concept

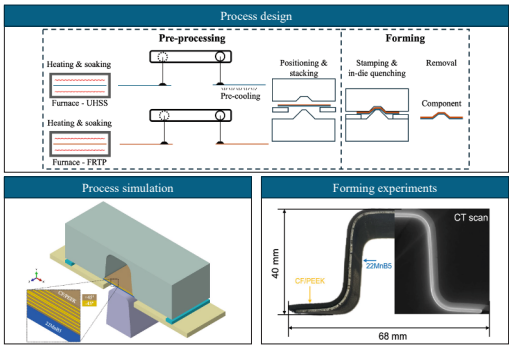
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HIGHLIGHTS

- A novel one-shot forming process to produce UHSS-based multi-material structures.
- A transient thermal-gradient strategy enables forming, bonding and hardening.
- An tailored integrated simulation scheme to guide the process window for CF/PEEK-22MnB5.
- 280 °C CF/PEEK and 520 °C 22MnB5 gave the preferred tested condition.
- Trials achieved good bonding, low delamination and >1400 MPa steel strength.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

Fibre–metal laminates (FMLs) combining fibre-reinforced polymer (FRP) with ultra-high-strength steel (UHSS) are promising for high-performance multi-material structures, but their manufacture is limited. The forming temperature mismatch commonly necessitates separate forming and adhesive bonding, increasing cycle time, tooling complexity and recycling difficulty. This study introduces a novel one-shot forming process for UHSS-based FMLs. Separately heated FRP and UHSS blanks are stacked immediately before cold-die forming; the pre-cooled austenitised steel locally melts the adjacent FRP surface, while the remaining laminate forms predominantly in the solid state. The process enables concurrent forming, composite-metal bonding, and steel quench hardening. A U-shaped CF/PEEK–22MnB5 demonstrator was designed to validate the feasibility. A two-step simulation framework guided process-window selection by predicting through-thickness temperature evolution during stacking and inter-ply damage during forming. Preferred stacking temperatures of 280 °C for CF/PEEK and 520 °C for 22MnB5 were identified, consistent with forming trials. Under these conditions, the demonstrator exhibited good composite-metal bonding, minimal delamination and a 22MnB5 hardness of 453 HV10, corresponding to an ultimate tensile strength above 1400 MPa. Dimensional assessment showed good geometrical accuracy, although residual springback/spring-in indicated scope for further optimisation. The results demonstrate the feasibility of rapidly manufacturing UHSS-based FML components for larger-scale adoption.

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1. Introduction

Driven by the increasingly rigorous energy efficiency standards, the transportation industry has turned to advanced lightweight materials and more aggressive engineering solutions [1,2]. The established use of lightweight metal alloys in vehicular structures, such as advanced high-strength steels (AHSSs), ultra-high-strength steels (UHSSs) and aluminium alloys [3,4], is increasingly being complemented by the emerging use of polymers and fibre-reinforced polymers (FRPs) [1]. To further reduce the structural weight, while maintaining cost-effectiveness, multi-material structures have drawn increasing industry interest.

Fibre metal laminates (FMLs) are a class of multi-material systems interlacing metal with FRP sheets. Combining the merits of both metals and FRPs, FMLs offer a high strength-to-weight ratio, excellent high fatigue resistance, high impact resistance, and other advantages [5,6]. Following their successful implementation in the aviation sector, the automotive sector has been exploring their potential in constructing multi-material structures, particularly in the body-in-white (BiW). Fig. 1 shows the components from real-world automotive applications and near-real-world research. A notable example is the multi-material B-pillar in BMW 7-series, the first known component integrated with UHSS boron steel and carbon fibre-reinforced polymer (CFRP) [7]. In addition, the LEIKA research project demonstrated that floor panels incorporating steel- and magnesium-based FMLs could achieve an additional 25 % weight reduction compared to their conventional full-metal counterparts [8]. These examples show the structural promise of FMLs, but they also highlight the need for manufacturing routes that are compatible with high-volume automotive production.

To facilitate the broader adoption of FMLs, various manufacturing processes have been developed to produce FML components. These include press-brake bending, hand or automatic lay-up, shot-peening forming, incremental sheet forming, laser forming and die forming [14]. Ding et al. [14] provided a comprehensive review on these technologies, and concluded that stamp forming is the most promising method for producing FML components in the automotive industry. This preference is primarily due to its capability to produce complex-shaped components with short cycle times, thereby enabling high-volume production required by the automotive sector.

Stamp forming of FMLs usually requires an elevated forming temperature for better formability, a process similar to metal hot stamping processes [15]. However, compared to hot stamping monolithic metal sheets, FML's multi-material nature introduces more complexity in the preparation steps, more specifically, thermal treatment, prior to the actual forming. Various pre-thermal treatment strategies have been developed in existing research. Mosse et al. [16] directly heated pre-fabricated polypropylene (PP)-based FMLs prior to forming. However, the prior FML fabrication process increases the overall production time and energy consumption, and this strategy is not viable for forming thermoset-based FMLs as thermoset polymers cannot be reshaped after curing. Behrens et al. [17] heated separately sourced metal and Polyamide 6 (PA6)-based FRP sheet stacks prior to forming. The bonding between the metal and FRP layers is established simultaneously as the component is shaped. Although viable for thermoplastic- [17] and thermoset-based FMLs [18], the lengthy in-die consolidation keeps production time and energy consumption similar to heating pre-fabricated FMLs. Guo et al. [19] separately heated steel and epoxy-based FRPs to their respective preferred forming temperatures before stacking and being stamped. This strategy leverages the thermo-mechanical behaviours of metals for an opportunity of improving the formability and as-formed strength of FML components. In addition, technologies that utilise compression molding [20] or resin transfer molding (RTM) [21] to produce FML components have also been developed. However, due to the significantly higher forming temperature required for hot stamping UHSS, these strategies cannot be directly adopted to produce UHSS-based FMLs.

Despite the variety of the pre-thermal treatment strategy, the forming temperature remains the most critical process parameter, particularly for thermoplastic-based FMLs. Fig. 2 summarises the forming temperature windows reported in existing literature, categorised by the thermoplastic matrices. A forming process may be described as melt-state forming when the forming temperature is above a thermoplastic's melting point (T_m), or solid-state forming when the forming temperature is between a thermoplastic's glass transition temperature (T_g)

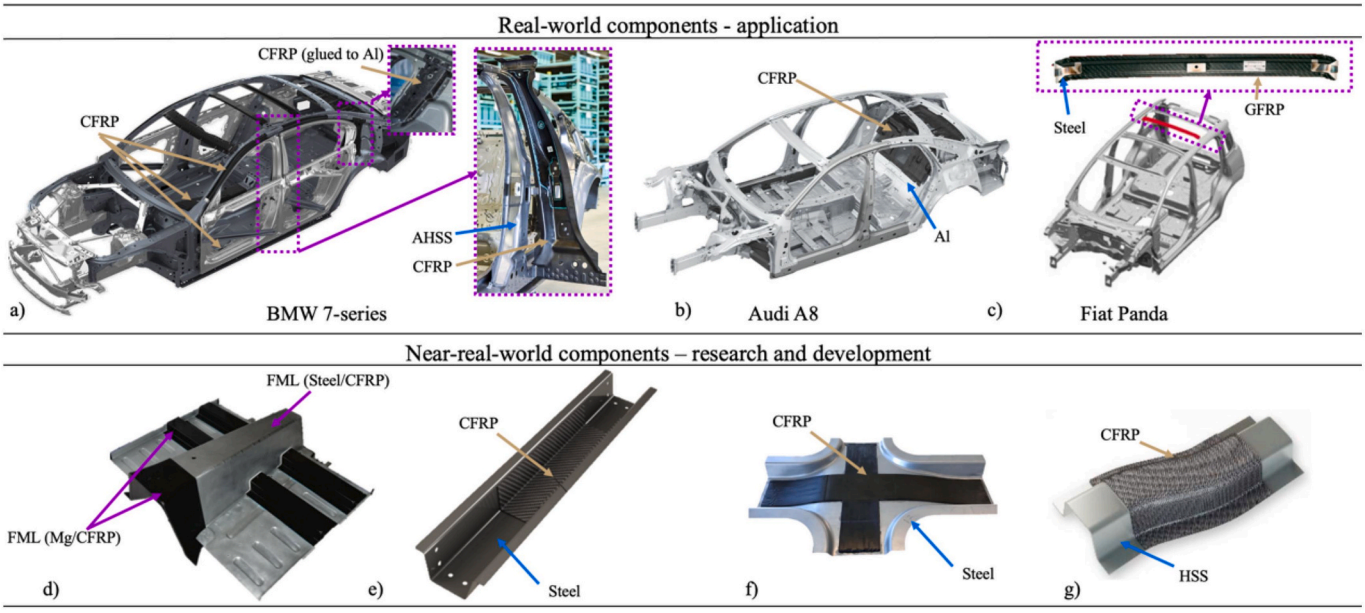


Fig. 1. Example of recent real-world fibre metal laminate (FML)/multi-material applications: a) the local carbon fibre-reinforced polymer (CFRP) on the C-pillars and ultra-high-strength steel (UHSS)-based FML B-pillar on BMW 7-series [7], b) the multi-material rear panel in Audi A8 [9], c) the glass fibre-reinforced polymer (GFRP)-reinforced roof rail on Fiat Panda [10]; examples of recent research development of near-real-world FML components: d) the LEIKA floor panel with steel- and magnesium-based FML components [8], e) a U-shaped pillar with local CFRP reinforcement [11], f) a steel-based cruciform demonstrator [12], and g) a high-strength steel (HSS)-based FML S-rail channel [13].

and T_m. It is found that the forming temperature varies but is mainly around the T_m and largely coinciding with the forming temperature of FRPs of the same matrix. Furthermore, while PP, PA6 and PVC are mostly used polymer matrices in existing research on FML forming, other advanced thermoplastics, such as PEEK, present an opportunity of constructing higher-performance FMLs.

Since the FML forming temperature is primarily determined by the polymer matrix, the preferable thermomechanical behaviours of metals are rarely considered. Most existing studies focused on metals suitable for cold-working, such as aluminium alloy 5505H34 [22], aluminium alloy AW-5754 [17], steel HC220Y [23], and steel DC04 [24]. While UHSS offers a significantly higher post-form strength, their extreme forming temperature of near 900 °C [25] will almost certainly damage the polymer matrix during forming in any existing one-shot FML forming technologies. Consequently, UHSS-based FML components, such as the previously mentioned B-pillars in the BMW 7-series [7], can currently only be produced by separately forming sub-components followed by secondary bonding. This method requires multiple sets of equipment and tools and complex multi-step processing, leading to higher tooling costs and higher manufacturing costs due to longer production time. In addition, the bonding between the metal and FRP layers often necessitates the use of adhesives, hindering the end-of-life recyclability of the as-formed components.

This study addresses the aforementioned process-window challenge by proposing a novel one-shot forming route for UHSS-based FMLs, demonstrated using a multi-material system constructed from a cross-plied unidirectional CF/PEEK laminate and a 22MnB5 steel sheet. The innovation lies in adding a tailored thermal sequence to a widely used steel hot-stamping and cold-die quenching setup to realise a new dynamic forming process: the CF/PEEK composite laminate and 22MnB5 steel sheet are heated separately, the austenitised steel is rapidly pre-cooled, and the blanks are stacked immediately before die closure so that the CF/PEEK surface in contact with the steel is locally melted while the remaining laminate forms in the solid state. This enables simultaneous deformation of the two constituents within their respective preferred temperature ranges, metal–composite bonding, and martensitic hardening of the steel within a single shot forming. The paper first develops the process concept and demonstrator design, then presents a stacking–forming simulation framework for process-window screening, and finally proves the feasibility of the new process and the selected conditions through forming trials and post-forming assessment.

2. Conceptual design of the one-shot forming process

Ultra-high-performance FML components require both constituents to retain their intended post-form performance. In the present material system, the 22MnB5 steel sheet must follow a thermal path that enables austenitisation, deformation before martensitic transformation and rapid quench hardening. By contrast, the CF/PEEK laminate must be heated sufficiently for forming and interfacial bonding, but must not be exposed to excessive temperatures or residence times that could degrade the PEEK matrix, or induce irreversible defects such as delamination and intra-ply splitting. The process design must therefore exploit the heat stored in the steel locally at the metal–composite interface while preserving the integrity of the remaining laminate. This section defines these process-window requirements, introduces the separate-heating, pre-cooling and one-shot forming strategy developed to satisfy them, and defines the demonstrator used to assess the concept.

2.1. Constituent thermal requirements and process windows

The proposed route should be governed by a transient process window defined by the competing thermal requirements of the UHSS sheet and thermoplastic composite laminate.

For 22MnB5, the steel must first be austenitised and then deformed and quenched at a sufficient cooling rate to obtain martensitic hardening [26]. Conventional hot stamping satisfies this requirement by transferring the austenitised blank to cold tools for forming and in-die quenching; however, typical corresponding steel temperatures of over 700–800 °C are far above the allowable thermal exposure of most polymer matrices. Direct contact between a conventionally hot-stamped steel blank and an FRP laminate would therefore risk matrix degradation, excessive melting and loss of laminate integrity.

Low-temperature (LT) hot stamping provides the steel-side opportunity for the proposed process. By introducing a controlled pre-cooling stage before deformation, the austenitised 22MnB5 blank can enter a lower forming temperature window while still achieving full martensitic hardening [27,28]. Successful LT hot stamping is achieved when the cooling rate exceeds the critical cooling rate and the deformation is completed before the blank cools below the martensitic transformation start temperature (M_s) [29]. A previous study [28] showed that an industry-level B-pillar demonstrator was successfully formed at temperatures as low as 500 °C. This significantly lower forming temperature window reduces the thermal severity of contact with the composite, while creating a short processing interval in which the steel remains hot enough to provide interfacial heat for bonding without damaging the composite. This offers an opportunity to form both materials in a single one-shot cycle.

On the composite side, solid-state stamp forming, where the composite laminates are formed between the glass transition temperature (T_g) and melting point (T_m), is attractive for UD-based laminates by utilising strain hardening to provide enhanced drawability [14]. In

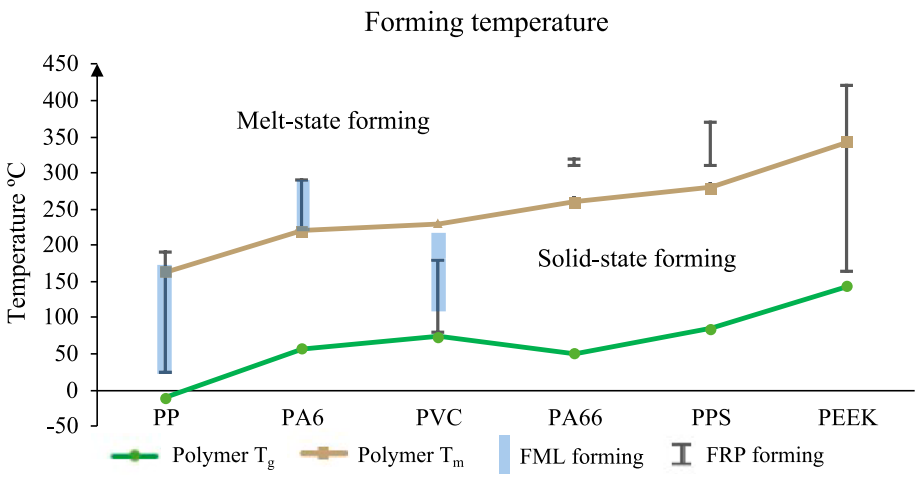


Fig. 2. Forming temperature windows of FMLs and FRPs with various thermoplastic polymer [14].

addition, as the fibres are constrained by the solid-state matrix from egressing the surfaces, better surface finishes can be achieved. Solid-state forming, compared to traditional melt-state thermoforming processes, also significantly reduces energy consumption and production cycle time.

For the present process, however, solid-state forming alone is insufficient: adhesive-free bonding to the steel requires local softening or melting of the CF/PEEK surface at the metal–composite interface. The desired thermal state is therefore deliberately non-uniform through the laminate thickness: a locally melted interfacial layer for bonding, combined with a predominantly solid-state laminate interior for controlled forming and preservation of post-form composite integrity.

2.2. Dynamic thermal-gradient strategy: separate-heating, pre-cooling, and one-shot stamping

To satisfy these coupled requirements, the proposed process uses a dynamic thermal-gradient strategy based on separate heating, controlled steel pre-cooling and immediate stacking before die closure. Fig. 3 illustrates the conceptual process design with a pre-processing strategy tailored for both UHSSs and FRPs. In this process, the UHSS and FRP blanks are heated and soaked separately. In particular, the UHSS needs to be completely austenitised during soaking. The austenitised UHSS is transferred, pre-cooled, and positioned on the tools. The FRP blank is then transferred and stacked on the pre-positioned UHSS. The stacked blanks are stamped into an FML component by closing the cold dies, followed by the in-die quenching.

Fig. 4 further illustrates the temperature profiles of the UHSS and FRP blanks undergoing this one-shot forming process, from which the key processing requirements are identified and detailed below.

Stacking. The stacking step is where the FRP is positioned onto the UHSS blank at their respective 'stacking temperature' for a short period of 'stacking time' for interfacial heat transfer. This crucial step creates a through-thickness temperature gradient within the FRP, where the outer surface of the FRP, which is in contact with the still hotter UHSS after pre-cooling, undergoes surface melting. The melted outer surface is cooled to solid-state again by the cold tools during stamping and in-die quenching, thereby establishing the bonding between the two materials. The inner layers of the FRP remain in the solid-state, providing the strain-hardening required for optimal formability.

UHSS critical cooling rate and M_s. UHSS temperature evolution must adhere to the LT hot stamping requirements, including not violating the critical cooling rate from A_3 and completing deformation before M_s for optimal formability and ultra-high post-form performance.

Removal temperature. The component can be released from the tools when the temperature of the FML, particularly the FRP, falls below its T_g.

These processing requirements are critical for achieving adhesive-free UHSS-FRP bonding and martensite formation in the UHSS, both of which are essential for successful forming of ultra-high performance UHSS-based FMLs. The cooling rate also affects the crystallinity and thus the stiffness and strength [30] of the as-formed FRP. However, its effects are not as significant compared to other requirements and therefore is deemed not the primary focus of this study.

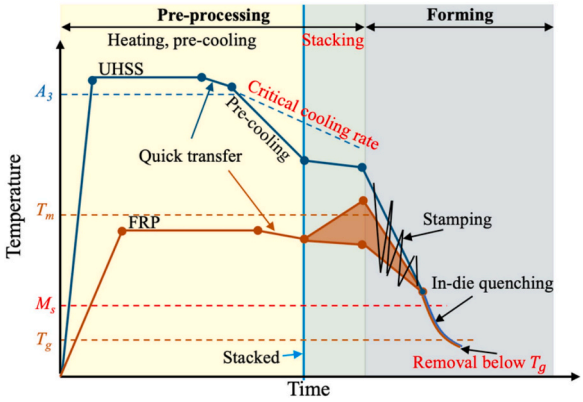


Fig. 4. Schematic illustration of the temperature profiles of UHSS and FRP during the forming process (red font and line denote the key processing requirements).

2.3. Demonstrator design

2.3.1. Material candidates

Carbon fibre-reinforced PEEK (CF/PEEK) laminate produced using cross-plied unidirectional prepregs and UHSS 22MnB5 are selected as the material candidates in this study. With the hot stamped 22MnB5 and CF/PEEK having ultimate tensile strength of 1500 MPa [25] and 2300 MPa [31], respectively, ultra-high performance UHSS-based FML can be produced. Moreover, using the boron steel 22MnB5 as the metal constituent of the FMLs poses the most challenging scenario to the proposed one-shot forming technology. The theoretical lowest forming temperature for 22MnB5 LT hot stamping is around 425 °C (M_s) [32], and the start forming temperature was recommended to be higher than 500 °C considering the heat loss to the cold die during forming [28]. A PEEK-based FRP is selected to accommodate these forming temperature requirements, with their T_g and T_m being 143 °C [33] and 343 °C [34], respectively, and decomposition onset temperature being 575 °C [35].

2.3.2. Component geometry

A U-bending demonstrator is designed as a proof-of-concept component for assessing the anticipated critical defects in this process, including bonding quality, delamination and springback. Fig. 5a, b) show the geometric design of the component, with a width of 40 mm and a drawing depth of 33.9 mm. The draft angle is 5°, thus the angle between the symmetric side walls and the top region is 95°. The component is designed with one layer of 22MnB5, 1 mm thickness, and one layer of cross-plied CF/PEEK laminate, 2.15 mm thickness and $[\pm 45]_{4s}$ layup. Fig. 5c) shows the blank size of 120 mm × 40 mm for both

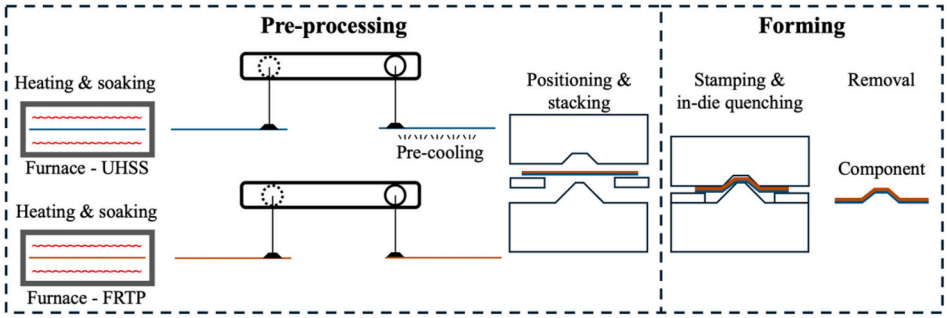


Fig. 3. Schematic illustration of the proposed separate-heating, pre-cooling and one-shot forming process.

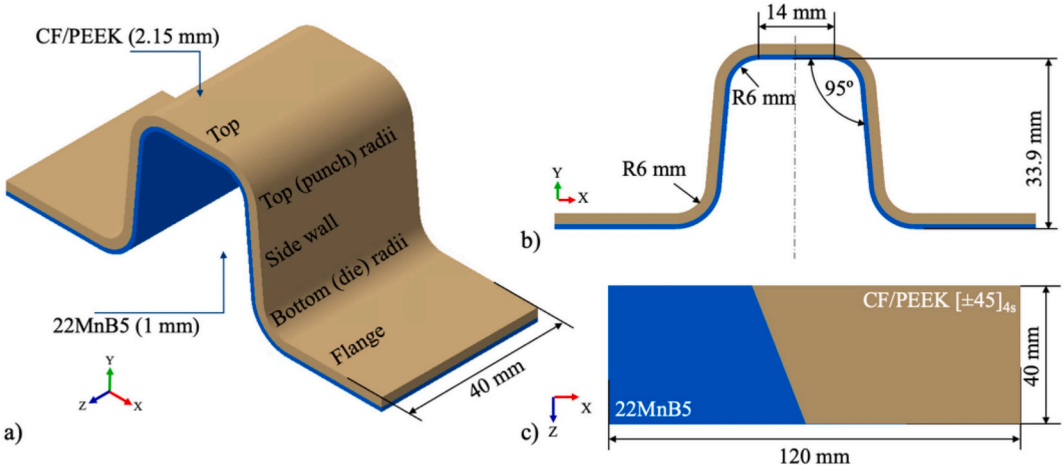


Fig. 5. Geometric design of the demonstrator component: a) isometric view of the component design, b) dimensions of the cross-section and c) blank size and CF/PEEK layup.

22MnB5 and CF/PEEK.

3. Process simulation

A finite element process-window study was developed to select stacking temperatures before forming trials. The objective was to identify temperature combinations that satisfy the key process requirements defined in Section 2: local CF/PEEK surface melting, acceptable inter-ply damage risk and retention of the 22MnB5 LT hot-stamping window.

The simulation scheme therefore consisted of two linked steps. The stacking step predicted the through-thickness temperature evolution after the separately heated CF/PEEK and 22MnB5 blanks were brought into contact. The forming step then used the propagated temperature field to estimate the relative risk of CF/PEEK inter-ply damage during tool closure. All simulations were conducted in Abaqus/Explicit.

3.1. Numerical model construction

3.1.1. Geometric modelling

Fig. 6 illustrates the model assembly, replicating the demonstrator design as detailed in Section 2.3. It consists of a set of tools, including die, punch, blank holder, and spacers, and the stacked workpiece. The set of tools, made of H13, were modelled as rigid bodies. The workpiece consists of a 2.2 mm thick cross-plied CF/PEEK laminate and a 1 mm thick 22MnB5 blank. The layup of the cross-plied CF/PEEK laminate was

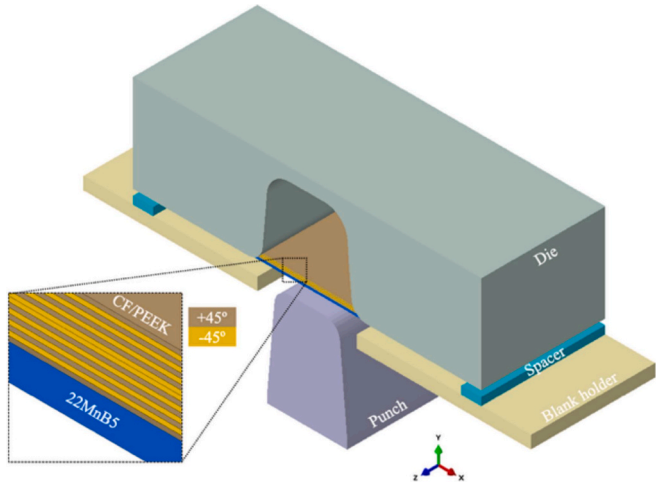


Fig. 6. Geometric modelling of the FML forming process.

$[\pm 45]_{4s}$, with the thickness of each ply being 0.1375 mm. Local material orientation was assigned to each ply, reflecting the layup configuration. The gap between the closed die and punch was 3.168 mm, slightly thinner than the combined blank thickness.

For the FEA meshing, each ply of the CF/PEEK was modelled with two layers, with the element size of all layers being 1 × 1 mm, and the 22MnB5 blank was modelled with three layers of deformable solid elements. The Coupled Temperature-Displacement element, C3D8RT, was chosen to capture the interfacial heat transfer during stacking and the material deformation during forming process.

3.1.2. Thermomechanical and thermal properties – CF/PEEK

The thermomechanical properties of each CF/PEEK ply were modelled with elastic–plastic behaviour at its preferred solid-state forming conditions.

Table 1 lists the elastic properties, including tensile moduli (E), shear moduli (G) and Poisson's ratio (ν). The subscripts 1, 2 and 3 represent the fibre longitudinal, in-plane transverse and out-of-plane transverse directions for each ply, respectively. The longitudinal modulus (E_{11}), dominated by the nearly inextensible fibres, was considered temperature-independent and obtained from material datasheet [31]. The temperature-dependent transverse tensile moduli E_{22} and longitudinal shear moduli (G_{12}, G_{13}) were determined by material models developed through in-house experiments. Poisson's ratios (ν_{12}, ν_{13}, ν_{23}) were calculated using the rule of mixtures (RoM) based on literature values for carbon fibre and PEEK [36] and the fibre volume fraction of 58.36 %. In addition, the material is assumed to be transversely isotropic.

The anisotropic plasticity of CF/PEEK was modelled with Abaqus's built-in Hill's potential function [37], taking the temperature-dependent longitudinal shear behaviour as the reference. The yield stress ratios, which need to be assigned in Abaqus, are defined in Eq. (1) [37],

Table 1
Elastic properties of CF/PEEK.

Temperature (°C)	E_{11}(GPa) *	E_{22}, E_{33}(MPa)**	ν_{12}, ν_{13}***	ν_{23}***	G_{12}, G_{13}(MPa)**
280	138	1470.35	0.354	0.166	1118.13
300		1069.52			789.83
320		668.69			461.54
343		207.74			84.01

*Material datasheet [31].
**Predicted using constitutive models developed through in-house experiments.
***Estimated using RoM model.

$$R_{11}=\frac{\overline{\sigma}_{11}}{\sigma^0},R_{22}=\frac{\overline{\sigma}_{22}}{\sigma^0},R_{33}=\frac{\overline{\sigma}_{33}}{\sigma^0},$$
$$R_{12}=\sqrt{3}\,\frac{\overline{\sigma}_{12}}{\sigma^0},R_{13}=\sqrt{3}\,\frac{\overline{\sigma}_{13}}{\sigma^0},R_{23}=\sqrt{3}\,\frac{\overline{\sigma}_{23}}{\sigma^0}, \quad (1)$$

where $\overline{\sigma_i}$ is the yield stress in a particular direction, and σ^0 is the reference yield stress, which is equal to $\overline{\sigma}_{12}$ in this study.

The reference longitudinal shear behaviour of the CF/PEEK was modelled using the constitutive model developed based on in-house experiments. The constitutive model, modified based on the classic Johnson-Cook model [38], is:

$$\sigma_s^f\left(\theta,\varepsilon_s^{pl}\right)=\left[\frac{\sigma_s^y}{\left(1-\widehat{\theta}^{m_s}\right)}+B_s\left(\varepsilon_s^{pl}\right)^{n_s}\right]\left(1-\widehat{\theta}^{m_s}\right)\lambda \quad (2)$$

where σ_s^f is the shear flow stress, and ε_s^{pl} is the shear plastic strain. σ_s^y is the yield strength, given by

$$\sigma_s^y(\theta)=\sigma_s^{ref}-\sigma_s^1\left(\theta-T_{ref}\right)^{m_s^y} \quad (3)$$

where σ_s^{ref} is the yield strength at the reference temperature T_{ref} of 280 °C, which is 4.347 MPa.

λ is a softening term, modified based on [39] by introducing the temperature dependency to improve the accuracy, and defined in Eq. (4):

$$\lambda=1-\varepsilon_s^{pl\left(\frac{k_s}{1-\widehat{\theta}^{m_s}}\right)} \quad (4)$$

$\widehat{\theta}$ is the nondimensional temperature defined as:

$$\widehat{\theta}=\begin{cases}0, & \theta<T_g\\ \left(\theta-T_g\right)/\left(T_m-T_g\right), & T_g\le\theta\le T_m\\ 0.99, & \theta>T_m\end{cases} \quad (5)$$

where T_g and T_m are the glass transition temperature and melting point of PEEK, respectively. It should be noted that when the current temperature (θ) is above the melting point (T_m), $\widehat{\theta}$ is not 1 as in the Johnson-Cook model but is modified to 0.99. This is to avoid the mathematical error of $\frac{\sigma_s^y}{\left(1-\widehat{\theta}^{m_s}\right)}$ becoming infinitely large.

σ_s^1, m_s^y, B_s, n_s, m_s and k_s are material constants for the plasticity model, and their values are listed in Table 2.

Fig. 7 shows the flow stress-plastic strain curves input into the simulation scheme. It should be noted that the stress–strain behaviours between 280 and 320 °C were interpolated while those between 320 and 343 °C were extrapolated based on the above-mentioned constitutive model. This was due to the difficulty of conducting characterisation experiments at temperature close to the T_m. An interval of 5 °C was deemed suitable for both interpolation and extrapolation, and the predicted flow stress-plastic strain curves were input to Abaqus using the lookup table option.

The ratio R_{11} was assigned as 100, a large enough value to ensure the linear response along the fibre direction [40], and was assumed temperature-independent. The ratios R_{12} and R_{13} were 1.732, also temperature-independent, as suggested by Eq. (1). The temperature-dependent transverse tensile ratios R_{22} and R_{33} were obtained by comparing the yield stresses between the transverse tension, which was predicted similarly as the longitudinal shear, and longitudinal shear at each temperature. The R_{23} was assumed to be the same as R_{22} and R_{33}

Table 2
Material constants for CF/PEEK's longitudinal shear plasticity model.

σ_s^1(MPa)	m_s^y	B_s(MPa)	n_s	m_s	k_s
0.189	0.612	499.76	0.64	0.43	0.18

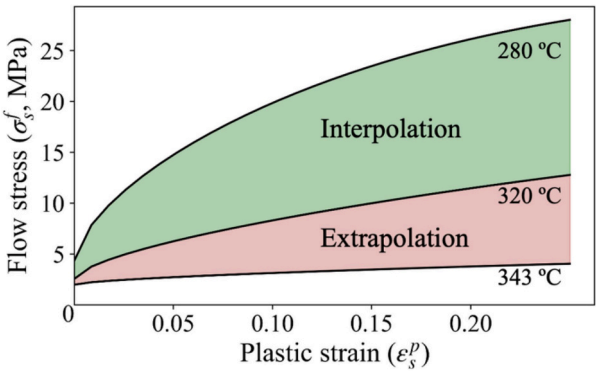


Fig. 7. Flow stress-plastic strain of CF/PEEK under the longitudinal shear deformation.

[37], as both transverse shear and transverse tensile deformation can be primarily attributed to polymer deformation and similar fibre-interface interactions in UD FRPs.

Table 3 lists other CF/PEEK properties, including density and thermal properties, used in the simulations. These properties were determined based on material datasheets using the RoM. In particular, the inverse RoM was used to estimate the thermal conductivity, as the primary heat transfer direction in this process is the through-thickness direction, which is perpendicular to the fibre direction.

3.1.3. Interfacial properties – CF/PEEK inter-ply cohesive behaviours

The interfaces between each CF/PEEK ply were modelled as cohesive surfaces using a cohesive zone model (CZM) to predict the risk of delamination. The behaviours of the cohesive surfaces were constituted by a bilinear traction-separation law [43]. In the traction-separation law, the surface stresses t_n, t_s and t_t increase linearly against the stiffness k_n, k_s and k_t, where the subscripts n, s, and t represent the normal and two shear directions to the cohesive surfaces, respectively. When one of the surface stresses reaches its maximum stress criterion, the surface damage initiates. This maximum nominal stress criterion can be expressed as Eq. (6),

$$\max\left\{\frac{\langle t_n\rangle}{t_n^0},\frac{t_s}{t_s^0},\frac{t_t}{t_t^0}\right\}=1 \quad (6)$$

where t_n^0, t_s^0 and t_t^0 are the damage initiation stresses, i.e., cohesive strengths, at respective directions. It should be noted that the Macaulay bracket is used for the normal traction so that negative traction is ignored as damage will not initiate under pure compressive stress.

Fig. 8 illustrates the temperature-dependent CZM used in this study, including both Mode I (normal direction) and Mode II (shear directions). The initial stiffness k_n, k_s and k_t were estimated using Eq. (7) [44] and (8) [45]:

$$k_n=\frac{\gamma E_3}{h_s} \quad (7)$$

$$k_s=k_t=k_n\frac{G_n^C}{G_s^C}\left(\frac{t_s^0}{t_n^0}\right)^2 \quad (8)$$

Table 3
Density and thermal properties of CF/PEEK [31,41,42].

	Density (kg/m^3)	Thermal conductivity (W/mK)	Specific heat (J/kgK)
CF	1800	10	710
PEEK	1300	0.25	2200
CF/PEEK (v_f: 58.36 %)	1591.80*	0.58**	1330.44*

*Estimated using the Voigt model (RoM).
**Estimated using the Reuss model (inverse RoM).

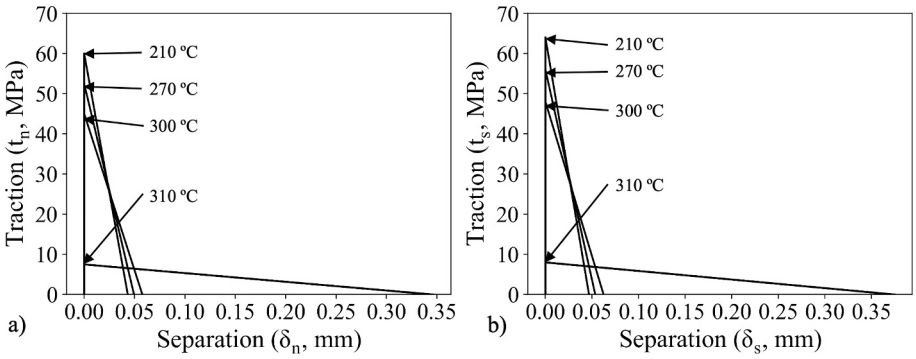


Fig. 8. Cohesive zone model (CZM) for a) Mode I (normal) and b) Mode II (shear).

E_3 is the through-thickness modulus, which was considered to be the temperature-dependent transverse tensile modulus as detailed in Section 3.1.2. γ is an empirical value and was set to 50 as literature [44] recommended. h_s is the sub-laminate (i.e., ply) thickness of 0.1375 mm. The normal (t_n^0) and two shear (t_s^0, t_t^0) cohesive strengths were estimated to be 75 % and 80 % of the PEEK strength, respectively [46,47]. In particular, the temperature-dependent PEEK strength was determined from a previous study [47]. However, due to the lack of readily available PEEK strength data above 310 °C and an effective predictive model, the cohesive strength was assumed constant above this temperature by Abaqus's default constant extrapolation method. G_n^C and G_s^C are the fracture energy, or fracture toughness, which equals to the area under the traction-separation curve, as shown in Fig. 8. The normal fracture toughness (G_n^C) was estimated to be 1.3 N/mm [48], and the shear fracture toughness (G_s^C) was estimated to be 1.5 N/mm, slightly higher than the normal fracture toughness.

After damage initiation, where the cohesive strength is reached, the Benzeggagh-Kenane (BK) mixed-mode fracture criterion [43] was used to govern the damage evolution. The material constant related to the BK model, η, was 2.3 [49]. Furthermore, a linear damage evolution was used, resulting in the cohesive surface damage, D, defined as in Eq. (9):

$$D = \frac{\delta_m^f\left(\delta_m^{\max} - \delta_m^o\right)}{\delta_m^{\max}\left(\delta_m^f - \delta_m^o\right)} \tag{9}$$

where $\delta_m^f = 2G^C/T_{eff}^o$ with T_{eff}^o as the effective traction at damage onset, $\delta_m^{\max}$ is the maximum effective displacement during the loading history, and δ_m^o is the effective displacement at damage initiation. D, is a scalar damage variable which monotonically evolves from 0 to 1 as the damage grows. Upon reaching 1, the interface fails and the adjacent plies delaminate [43].

3.1.4. Thermomechanical and thermal properties – 22MnB5

The thermomechanical properties of the 22MnB5 were modelled as isotropic elastic-viscoplastic. Based on the conceptual process design, the 22MnB5 forming temperature would be around 500 °C. Therefore, the thermomechanical properties within the temperature range between 400 °C and 600 °C were used in the simulation. The temperature-dependent elastic properties are listed in Table 4, along with other thermal properties.

Fig. 9 illustrates the temperature-dependent viscoplastic behaviours, adopted from PAM-STAMP 2019 material database [51].

Table 4
Material properties of 22MnB5 [50,51].

Temperature (°C)	Density (kg/m^3)	Elastic modulus (GPa)	Poisson's ratio	Thermal conductivity (W/mK)	Specific heat (J/kgK)
400		166	0.298	21.7	561
500	7800	158	0.303	22.65*	573
600		150	0.310	23.6	581

*Abaqus interpolation.

3.1.5. Frictional behaviours

After debonding, the CF/PEEK plies start sliding against each other under the frictional law, with a friction coefficient of 0.4 [52]. In addition, Table 5 lists the pressure-dependent friction coefficients of CF/PEEK-22MnB5 and CF/PEEK-tool interfaces. These friction coefficients were adopted from existing research on other FRPs as an approximation [53–55]. The friction coefficient between the tool and 22MnB5 was 0.68 according to literature [56].

3.2. Simulation scheme and boundary conditions

Forming temperature is the most critical process parameter in both FRP solid-state forming and UHSS hot stamping. However, as a complex through-thickness temperature distribution is expected in the forming process, the stacking temperatures were used as the input variables and assigned to blanks uniformly as a pre-defined field, followed by a two-step simulation scheme. The two-step simulation scheme was developed to reflect the physical forming process, and begins with stacking, where the CF/PEEK was heated by the 22MnB5, followed by forming. Table 6 lists the FE model types and parameters for both steps, which are detailed in the following two sections. The mass scaling factors were deemed suitable through trial and error.

The simulations were conducted with various stacking temperature combinations, considering the requirements for both materials. Wang et al. [47] suggested that the PEEK-based FRP should be formed around 310 °C. However, as the CF/PEEK is expected to be heated during stacking, lower stacking temperatures were designed. Ganapathy et al. [28] suggested that the forming temperature of 22MnB5 should be higher than 500 °C. Therefore, the simulations were conducted at stacking temperature ranges between 260–300 °C and 500–540 °C for

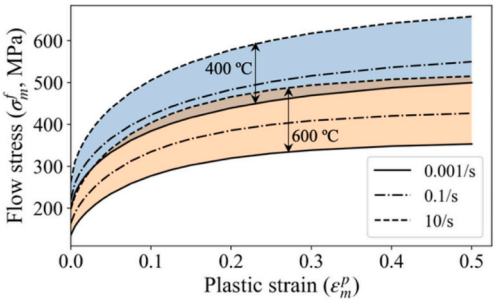


Fig. 9. Flow stress-plastic strain of the 22MnB5 covering simulated forming temperatures (adopted from material database of PAM-STAMP [51], σ_m^f and ε_m^p denote the flow stress and plastic strain of the metal, respectively).

Table 5
Friction coefficients of CF/PEEK-22MnB5 and CF/PEEK-Tool interface [53,54].

Pressure (MPa)	CF/PEEK-22MnB5	CF/PEEK-Tool
0.02	1.25	–
0.04	0.75	–
0.1	0.3	0.4
0.15	0.2	–
0.5	–	0.2

Table 6
FE simulation scheme and model parameters.

	Step 1	Step 2
Description	Stacking	Forming
Type	Dynamic, Temp-disp, Explicit	Dynamic, Explicit
Step time	1.5 s	0.36232 s
Mass scaling	1E6	5E4

CF/PEEK and 22MnB5, respectively, with the test matrix as listed in Table 7.

3.2.1. Step 1: Stacking

In the stacking step, the CF/PEEK and 22MnB5 blanks, at their respective pre-defined stacking temperatures, were assumed to be an adiabatic system, considering only the interfacial heat transfer between the two materials and neglecting heat loss by convection to the ambient or conduction to the tools. The interfacial heat transfer coefficient was 1.5 kW/m^{2}K, which was obtained from an in-house experiment. In addition, as CF/PEEK was modelled as individual plies sequentially in contact, instead of an integral laminate, an artificial interfacial heat transfer coefficient is required to enable the heat conductance. However, this approach reduces the heat transfer efficiency compared to modelling a continuous laminate with the thermal conductance. To minimise the impact, a relatively large interfacial heat transfer coefficient of 1000 kW/m^{2}K was used.

In-house experiments suggested the stacking time should be less than 2 s to avoid early phase transformation in the 22MnB5. Therefore, a stacking time of 1.5 s was simulated. The resultant through-thickness temperature gradient was then propagated to the forming step.

3.2.2. Step 2: Forming

Fig. 10 illustrates the boundary conditions of the forming step, which reflect the physical forming process. The punch was fixed in all 6 degrees of freedom (DoFs), while other tools, including blank holder, spacers and die, were fixed in all but the Y-direction translation, enabling the vertical displacement. The die, as the driving part, was displaced 36.232 mm in the negative Y-direction until the tools were fully closed within the step time, at a forming speed of 100 mm/s. Therefore, the step time was set to be 0.36232 s. A total blank holding force of 60 kN was uniformly applied to confine the blanks within the gap created by the spacers. The 22MnB5 and CF/PEEK blanks, unconstrained in any DoFs, were deformed as the tools closed.

As the forming time was much shorter compared to the stacking time and being only ~0.36 s, the blank temperature change during step 2 is limited, especially for CF/PEEK considering their relatively low through-thickness thermal conductivity. Therefore, the forming step is simplified as isothermal. The thermal expansion, crystallisation and evolving interfacial properties, especially cohesive behaviour between CF/PEEK plies, are not captured in the present model, limiting its capability or accuracy in predicting post-form geometrical accuracy and interfacial quality. However, as a process window screening tool, this simulation scheme has shown adequateness in achieving the objective of identifying preferred forming temperatures.

Table 7
Test matrix for simulation investigating the effects of stacking temperatures.

22MnB5 (°C)	CF/PEEK (°C)		
	260	280	300
500	√	√	√
520		√	
540		√	

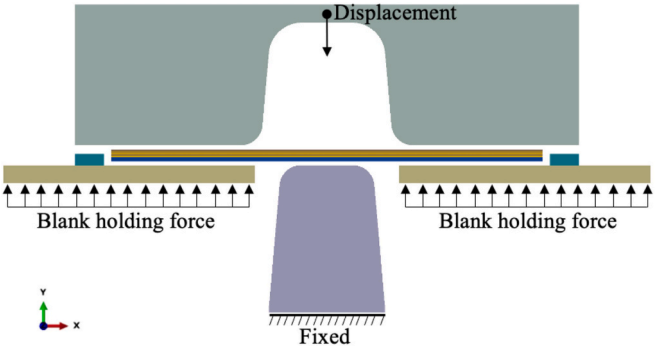


Fig. 10. Boundary conditions for forming.

3.3. Simulation result analysis

3.3.1. Temperature field evolution in the stacking step

As the stacking is a critical step in the proposed forming process, where the surface of the CF/PEEK is expected to be melted by the 22MnB5 through interfacial heat transfer to facilitate the bonding, results from this step are first analysed.

Fig. 11 shows the temperature evolution of the CF/PEEK-22MnB5 interface under various stacking temperature combinations within a stacking time of 1.5 s. Fig. 11a) and b) show the surface temperatures of 22MnB5 remained above the M_s of around 425 °C at all stacking temperature combination after stacking. In particular, stacking the 22MnB5 and CF/PEEK at 500 °C and 260 °C, respectively, leading to the lowest temperature of 455 °C in the 22MnB5. This suggests that the LT hot stamping requirements were satisfied in all simulated conditions by keeping the 22MnB5 above M_s prior to the forming commences. Fig. 11c) and d) show the surface of CF/PEEK was heated by the 22MnB5 to above not only its T_m, but also the recommended minimum consolidation temperature at all stacking temperature combinations. While the cooling during the forming and in-die quenching was not simulated, the melted PEEK matrix will be reconsolidated and act as adhesive allowing the post-form bonding between the CF/PEEK and 22MnB5.

Furthermore, Fig. 12 shows the through-thickness temperature gradients after stacking for 1.5 s, with at least the first three plies closest to the 22MnB5 expected to be melted during stacking. However, instead of modelling the CF/PEEK laminate with sharing-nodes in the ply interfaces, where the contacting surfaces of adjacent plies should have identical temperature, a temperature difference exists between contacting ply surfaces. This can be observed in Fig. 12 by comparing temperature ranges, represented by the error bars, of adjacent plies. These temperature differences are attributed to the reduced heat transfer efficiency, despite using a large interfacial heat transfer coefficient, and are especially large in plies closer to the 22MnB5.

This reduced efficiency in inter-ply heat transfer leads to an overestimation of temperatures in plies closer to the 22MnB5 and an underestimation in the plies further away. At the lowest stacking temperatures of 260 °C for CF/PEEK and 500 °C for 22MnB5, only one ply exceeded the consolidation temperature, resulting in a narrow forming window.

3.3.2. Cohesive surface damage in the forming step

The blanks, at their final temperatures after stacking for 1.5 s, were

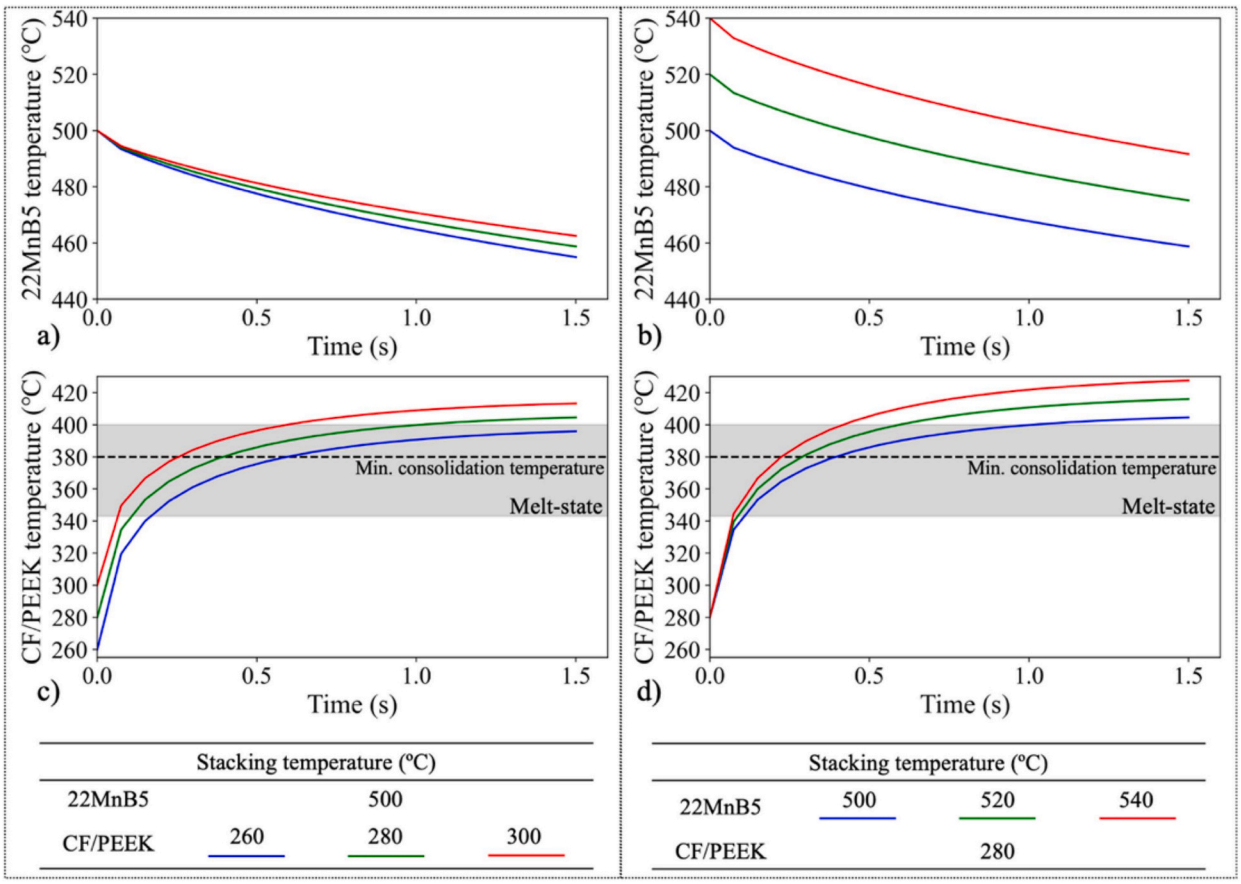


Fig. 11. Surface temperature evolution of a, b) 22MnB5 and c, d) CF/PEEK under various stacking temperature combinations.

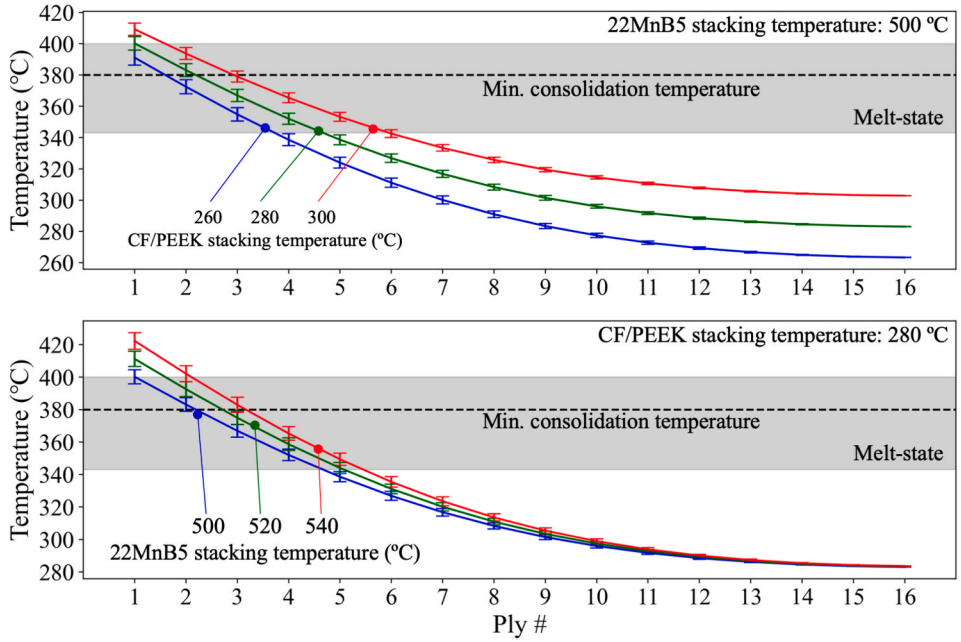


Fig. 12. Through-thickness temperature gradient of the CF/PEEK after stacking for 1.5 s at different stacking temperature combinations (Ply #1 is closest to the 22MnB5, the upper and lower error bars represent the temperatures of surfaces closer to and away from the 22MnB5, respectively).

then formed by the closing tools. The cohesive surface damage, D, in ply interfaces of the CF/PEEK was analysed to evaluate the delamination risk during forming.

Fig. 13 and Fig. 14 show the effect of stacking temperatures on D of each ply interface. While the interfaces fail and debond when D reaches 1 by the definition of CZM, the tendency of failure occurrence is considered high when D is 0.99 in this study. Therefore, the D contour values below 0.99 were filtered out for visual clarity. A noteworthy

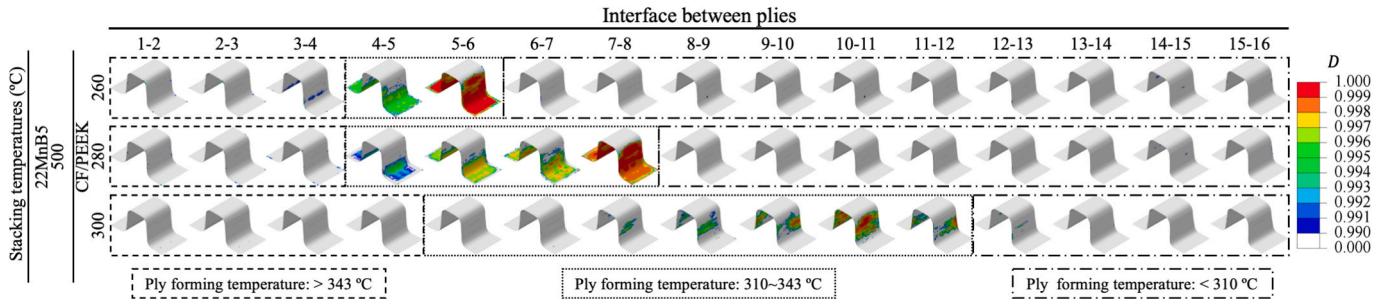


Fig. 13. Cohesive surface damage (D) field of each ply interface within the formed CF/PEEK: effect of CF/PEEK stacking temperatures with 22MnB5 stacking temperature being 500 °C (Ply #1 is closest to the 22MnB5).

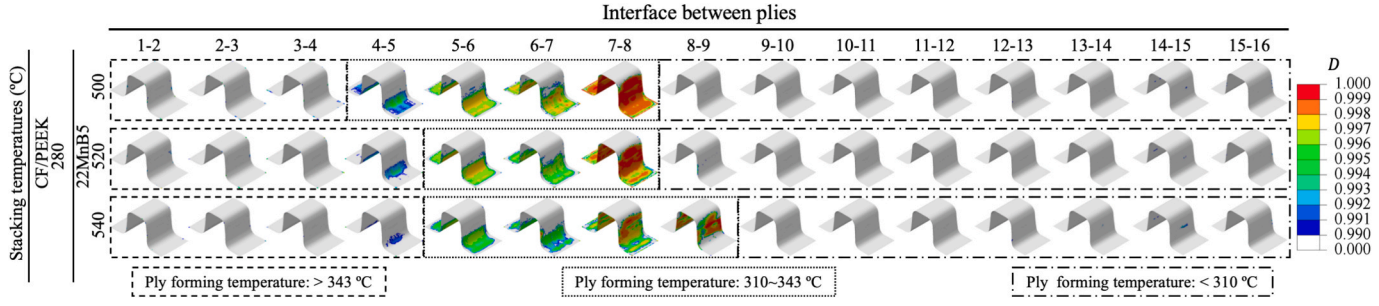


Fig. 14. Cohesive surface damage (D) field of each ply interface within the formed CF/PEEK: effect of 22MnB5 stacking temperatures with CF/PEEK stacking temperature being 280 °C (Ply #1 is closest to the 22MnB5).

observation is that the plies with forming temperatures between 310 and 343 °C, although all D are lower than 1, have considerable large regions at risk of delamination. This phenomenon could be caused by a stiffness mismatch between the cohesive surfaces and their adjacent CF/PEEK plies, due to the lack of accurate cohesive properties near the T_m. In further analysis, the plies were, therefore, grouped by their forming temperatures into above T_m (343 °C), between 310 and 343 °C, and below 310 °C.

The plies with forming temperatures higher than 343 °C are almost defect free, except at higher 22MnB5 stacking temperatures (520 and 540 °C), in which cases the interface between Ply #4 and #5 shows relatively larger delamination risk at the bottom (die) radii and flanges. Nevertheless, since plies in this group are expected to melt, delaminated plies can be re-consolidated when the tools close with pressure. Re-consolidation could also occur within plies with forming temperatures between 310 and 343 °C, as the actual ply temperature could be closer to or higher than T_m due to the heat transfer efficiency previously discussed. Therefore, the delamination within this group could also be remedied. Plies with forming temperatures below 310 °C, however, can hardly be melted. Therefore, delamination within this group is irreversible and of particular interest.

To further analyse the delamination risk quantitatively, the maximum cohesive surface damage (D_{max}) of each ply interface was investigated. Fig. 15a) shows the effects of increasing the CF/PEEK stacking temperature with a constant 22MnB5 stacking temperature of 500 °C. While increasing the CF/PEEK stacking temperature from 260 °C to 280 °C reduces the D_{max} overall and in the below 310 °C ply interface group, further increasing the stacking temperature to 300 °C increases the delamination risk. These non-monotonic phenomena are the effect of the interactions of the temperature-dependent material behaviours and CZM of the CF/PEEK. As the cohesive strength is lower at higher temperature and fracture energy is assumed constant, larger separation is allowed between adjacent plies, reducing the delamination risk. However, increasing the forming temperature also makes the plies more easily being deformed hence the D_{max} is greater when the stacking temperature is increased to 300 °C. This can be further confirmed by investigating the effects of increasing the 22MnB5 stacking temperature with the CF/PEEK stacking temperature being 280 °C.

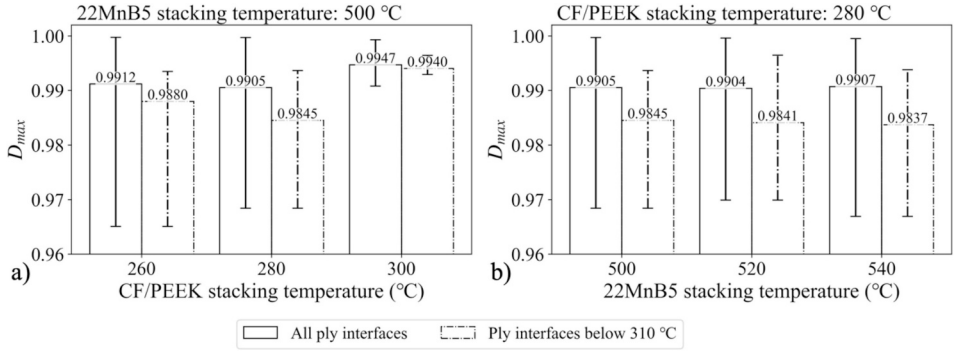


Fig. 15. Effect of a) CF/PEEK stacking temperature with a 22MnB5 stacking temperature of 500 °C and b) 22MnB5 stacking temperature with a CF/PEEK stacking temperature of 280 °C on the D_{max} of ply interfaces (bars and error bars represent the average values and ranges, respectively, of D_{max} in ply interfaces within corresponding group.

Fig. 15b) shows a similar non-monotonic trend of the all-ply average D_{max} as the 22MnB5 stacking temperature increases, while the average D_{max} continuously decreases in ply interfaces below 310 °C. As Fig. 14 suggests, more plies are formed at higher temperatures of above both 310 and 343 °C when the 22MnB5 stacking temperature is 540 °C, leading to higher D_{max} in these ply interfaces thus the average D_{max} of all ply interfaces.

The cohesive damage analysis suggests that the preferred stacking temperatures are 280 °C and 520 °C for CF/PEEK and 22MnB5, respectively. The effects of stacking temperatures are further investigated through forming trial experiments, as detailed in the next section.

4. Forming trial experiments and post-form assessments

4.1. Experimental design

4.1.1. Specimen materials

The boron steel used in the forming trials was Al-Si coated 22MnB5, with a thickness of 1 mm, provided by Shougang Group. Table 8 lists the chemical composition of the material. The FRP was in-house produced using the prepregs (Teijin TPUD [31]) and a hot press. The layup was $[\pm 45°]_{4s}$ and the thickness was 2.15 ± 0.1 mm. The 22MnB5 and CF/PEEK blanks, as designed in Section 2.3.2, were cut by wire cutting and waterjet cutting, respectively.

4.1.2. Forming tools design and experiment setup

Fig. 16 shows the design of the forming tool set used to form the demonstrator component. The tool set consists of a die, punch, blank holder, and spacers. The dimensions of the punch surface are identical to the lower surface of the designed component. The gap between the die and punch was controlled by curvature offsetting for the top and side wall regions and spacers for the flange regions.

It should be noted that this gap should be slightly larger than the total blank thickness in experiments, instead of 1 % thinner as used in the simulations. Unlike numerical simulation, where the blanks were assumed to have exact designed thickness before the forming starts, real-world materials suffer from thickness variation due to production accuracy and thermal expansion at elevated temperatures. To ensure an adequate CF/PEEK-22MnB5 bonding and avoid over-squeezing the CF/PEEK, initial experiments suggested that the ideal gap should be between 3.3 and 4.4 % thicker than the total blank thickness [57].

Moreover, a ‘hard stop’ was designed on the punch to avoid over-squeezing the FRPs when the tool closes. The blanks are drawn into the die by closing the tools.

Two sets of pins were specially designed in the blank holder. The holding pins support the steel blank during pre-cooling and stacking, avoiding excessive uncontrolled cooling by the cold blank holder, thereby preventing premature microstructural phase transformation. The position pins, slightly higher than the holding pins, serve as “stoppers”, so that the blanks can be precisely placed and stacked before forming. Both sets of pins are supported by springs, so that they can move downwards when the tool closes.

Fig. 17a) shows the forming tool set being assembled onto an in-house developed integrated forming system [57,58]. The integrated forming system comprises a mechanical die frame with four 15 kN gas springs supporting the middle plate and mechanical springs supporting the top plate. A set of air nozzles, which are connected to the 100 psi infrastructure compressed air through a series of solenoid valves controlled by a compact NI-RIO controller, are assembled on the top plate. The forming tool assembly was then placed onto the ESH 25T high-rate machine. Fig. 17b) shows the experiment setup for the forming trials. Two furnaces were placed on each side of the machine for heating the UHSS and FRP, respectively.

4.1.3. Experiment procedure

Fig. 18 shows the experimental procedure in practice with designed timeline for the forming trials, following the conceptual pre-processing strategy outlined in Section 2.2. Prior to the forming process, the CF/PEEK and the 22MnB5 blanks were heated and soaked in the corresponding furnace for 15 mins and 3 mins, respectively. When the furnace for the 22MnB5 opened at t0, the blank was transferred onto the holding pins using a pair of tongs, followed by an 8 s active air pre-cooling time. During the pre-cooling, the CF/PEEK blank was retrieved from its furnace using the same tongs and, once the pre-cooling was complete, transferred on top of the 22MnB5 and stacked. The stacked blanks were then formed by closing the tools.

It should be noted that this experimental procedure was developed under laboratory environment. The relatively long heating and soaking time was to ensure the blanks are uniformly heated and stabilised and was determined by independent heating trials. In industrial production, the heating time can be significantly reduced by heating multiple blanks or utilising advanced industry-level heating technologies and equipment.

4.1.4. Test matrix

The effects of stacking temperatures, as the most critical process parameters, were investigated. Table 9 lists the test matrix of the target stacking temperatures and trial numbers. The conditions are identical to those simulated in Section 3, as the simulation results would provide insights in understanding the physical phenomenon observed in experiments, and complementarily, the direction of potential modelling improvements can be identified through the experiments.

Other forming process parameters were kept consistent for all forming trials. Specifically, the blank holding force, forming speed and holding time were 60 kN, 100 mm/s and 30 s, respectively. In particular, the relatively long holding time of 30 s was used to ensure the key process requirement of removal below T_g was satisfied.

4.2. Forming trial experiments and result analysis

4.2.1. Actual stacking temperatures

Prior to each forming trial, the temperature profiles for both 22MnB5 and CF/PEEK were measured. Fig. 19 illustrates the design of the specimen, both 22MnB5 and CF/PEEK, used for temperature measurements. A thermocouple was inserted into the pre-drilled hole as shown in the figure.

Fig. 20a) shows the example temperature profiles of the 22MnB5, with an 8 s active air cooling time. The stacking temperatures of 500 and 520 °C were achieved by adjusting the valve opening to obtain different cooling rates, while maintaining soaking temperature near 950 °C. However, further reducing the cooling rate for a stacking temperature of 540 °C could violate the critical cooling rate. To avoid the undesired premature phase transformation caused by an insufficient cooling rate, a higher soaking temperature of 990 °C was used to achieve the stacking temperature of 540 °C.

As the CF/PEEK is subject to natural cooling during transfer, designed stacking temperatures can be achieved by simply adjusting the soaking temperature. Fig. 20b) shows the example temperature profiles during transferring the CF/PEEK blanks, measured using a temperature measuring specimen, with the same design as the 22MnB5, but a

Table 8
Chemical composition of 22MnB5 provided by Shougang Group in weight percentage.

Element	C	Si	Mn	P	S	Cu	Cr	Al	Ti	Mo	B
22MnB5	0.213	0.252	1.21	0.011	0.002	0.014	0.203	0.041	0.037	0.007	0.003

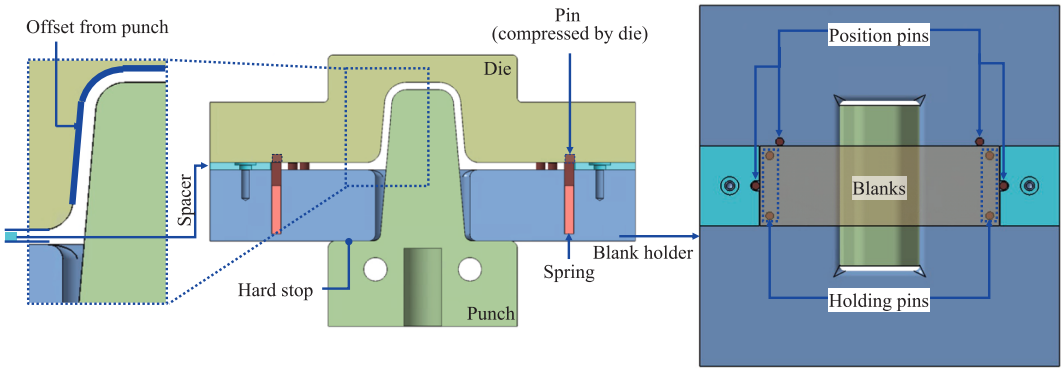


Fig. 16. Forming tool design.

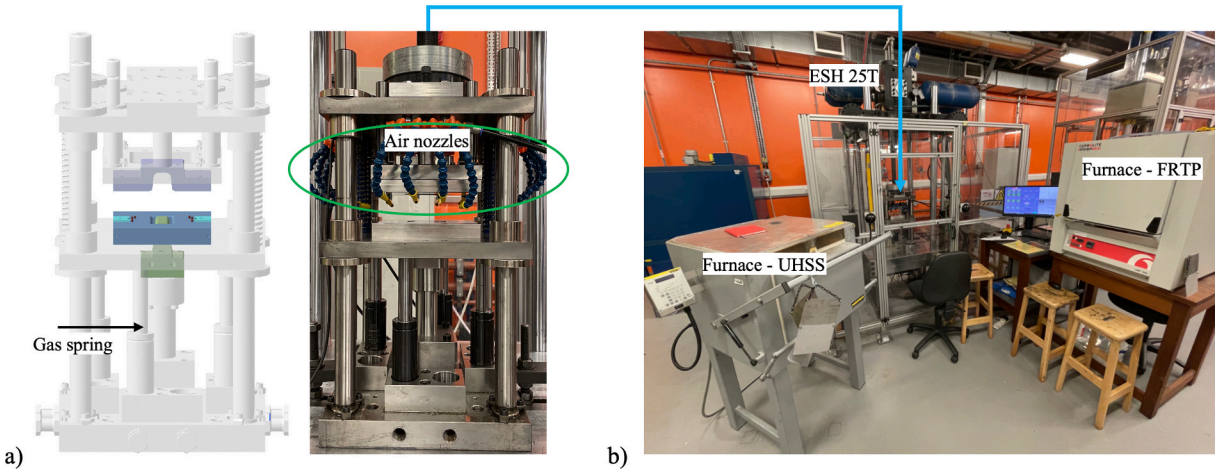


Fig. 17. a) Forming tool assembly (cad and picture) and b) forming trial experiment setup.

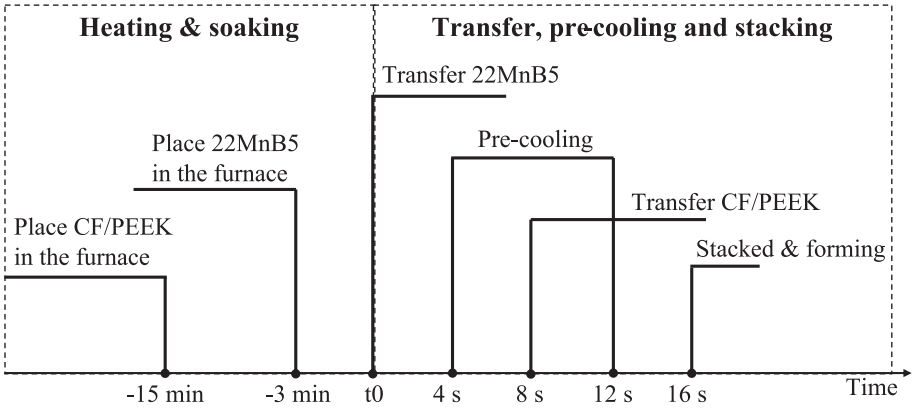


Fig. 18. Timeline of the experimental procedure for the forming trials.

Table 9
Test matrix of target stacking temperatures and trial numbers.

22MnB5 (°C)	CF/PEEK (°C)		
	260	280	300
500	①	②	③
520		④	
540		⑤	

thickness of 2.15 mm.

For both 22MnB5 and CF/PEEK, the measured temperatures at the designed stacked time (16 s after opening the 22MnB5 furnace) were used to calibrate process parameters, including the soaking temperatures and valve opening, within a tolerance of ±10 °C. However, transferring the material manually limited the temporal precision of the process, leading to variations in the stacked time, thereby affecting the stacking temperatures. To determine the actual stacking temperatures, the forming trials were recorded to obtain the actual time of each step, based on which the relevant temperatures were extracted from the temperature profiles measured before the forming trials. The actual cooling rates for the 22MnB5 from A_3, the austenitisation completion

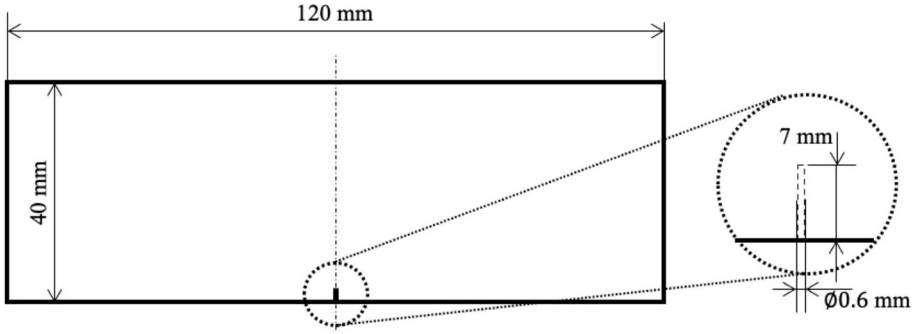


Fig. 19. CAD illustration of the specimen for temperature measurements.

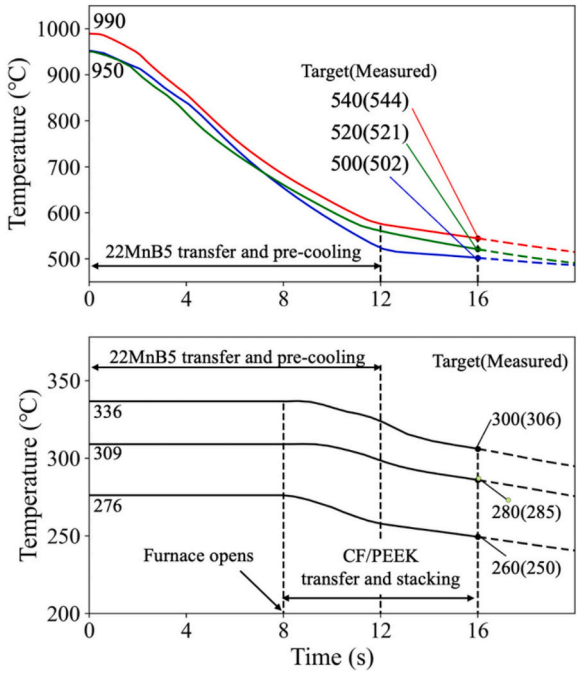


Fig. 20. Example temperature profiles of a) 22MnB5 and b) CF/PEEK measured prior to forming trials (dash lines represent continuous natural air cooling after the designed stacking time).

temperature, were also calculated to ensure the critical cooling rate requirement was not violated. Table 10 lists the actual stacked time, temperatures, and cooling rates for each forming trial. As the table shows, both stacking times and resulting temperatures as well as 22MnB5 cooling rates show variations due to manual material handling, while in industrial production where existing automated hot stamping lines are readily available to be upgraded for this process for a robust thermal control.

Table 10
Forming process parameters of the forming trials.

Trial number	Soaking Temperatures (°C)		Stacked time (s)	Stacking Temperature Target (Actual) (°C)		22MnB5 cooling rate from A_3 (°C/s)
	CF/PEEK	22MnB5		CF/PEEK	22MnB5	
①	281	950	15.6	260 (253)	500 (496)	30.3
②	309	963	13.8	280 (289)	500 (494)	35.6
③	337	951	17.2	300 (304)	500 (497)	27.4
④	308	951	14.9	280 (280)	520 (526)	29.5
⑤	308	989	14.7	280 (281)	540 (555)	29.3

4.2.2. *Effects of stacking temperatures*

Appearance by visual inspection

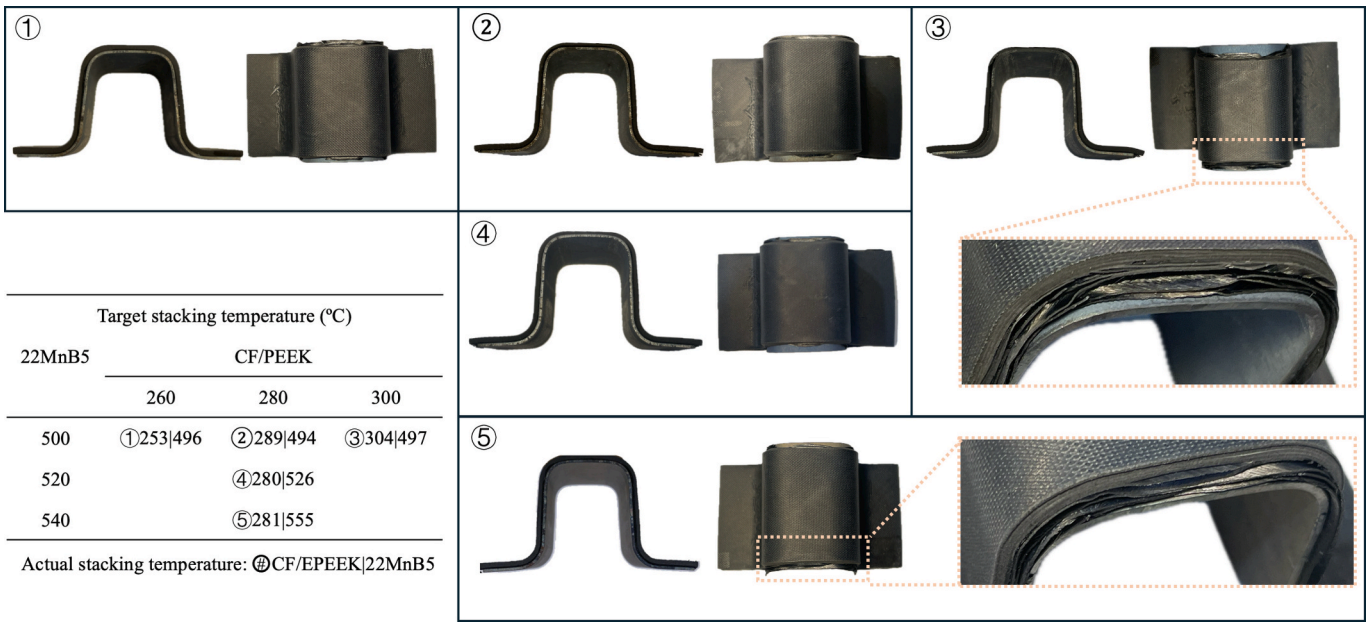
Fig. 21 shows the appearance of as-formed FML components demonstrating the effects of the stacking temperatures. It is evident that increasing the stacking temperature of either material constituent led to the CF/PEEK having high forming temperatures, resulting in bias-extension-like necking in the middle section under stretch in trials ③ and ⑤.

Interfacial quality by computed tomography (CT) inspection

The interfacial qualities between the CF/PEEK and 22MnB5, and the CF/PEEK plies, as crucial merits when evaluating the quality of the as-formed FML components, were examined by computed tomography (CT) with a scanning resolution of 50 μm. Fig. 22 shows the CT scan results, reviewed using a data viewer, myVGL 2023.4.

The CF/PEEK-22MnB5 interlaminar bonding was attained in all forming trials except for trial ①. In the cross-sectional views of Trials ③ and ④, the bonded interfaces appear continuous, with no obvious gap between the two materials. By contrast, Trial ① shows a clear sharp gap between the CF/PEEK laminate and 22MnB5 sheet in the top radii, indicating that bonding was not established at this location. As trial ① used the lowest stacking temperature combination, previous simulation suggests this may cause the surface of the CF/PEEK not to be sufficiently heated by the 22MnB5, potentially causing inadequate bonding. This risk was even higher in the forming trial due to the actual CF/PEEK stacking temperature (253 °C) being lower than the target (260 °C), as detailed in Table 10. Moreover, the CT scan shows that the no-bonding region is in the top (punch) radii. This localised lack of bonding can be attributed to the 22MnB5 blank being pressed against the punch radii, the first contact point once the tools started to close, and rapidly cooled. Due to the more efficient interfacial heat transfer between the punch and 22MnB5 blank compared to the CF/PEEK-22MnB5 interface, less heat is available to melt the CF/PEEK surface, thereby causing the inadequate bonding in these regions. It should be noted that while observing interface continuity can qualitatively confirm the establishment of CF/PEEK-22MnB5 bonding, evaluating the bonding quality demands further quantitative mechanical tests. Tests, such as lap shear, peel or double cantilever beam (DCB), are required in future work to assess interfacial strength and failure modes under different load bearing conditions.

Inter-ply delamination in the CF/PEEK was identified and marked in the CT scan images and displayed in 3D reconstruction for all as-formed components, with the image marking of trial ③ as an example. Delamination primarily occurred in three areas: the side walls, top (including the top radii), and bottom (die) radii regions. Delamination in the side walls can be observed in trials ① and ②, where the stacking



Target stacking temperature (°C)			
22MnB5	CF/PEEK		
	260	280	300
500	①253|496	②289|494	③304|497
520		④280|526	
540		⑤281|555	

Actual stacking temperature: ⓐCF/EPEEK|22MnB5

Fig. 21. Effects of stacking temperature on the formed FML components: appearance by visual inspection.

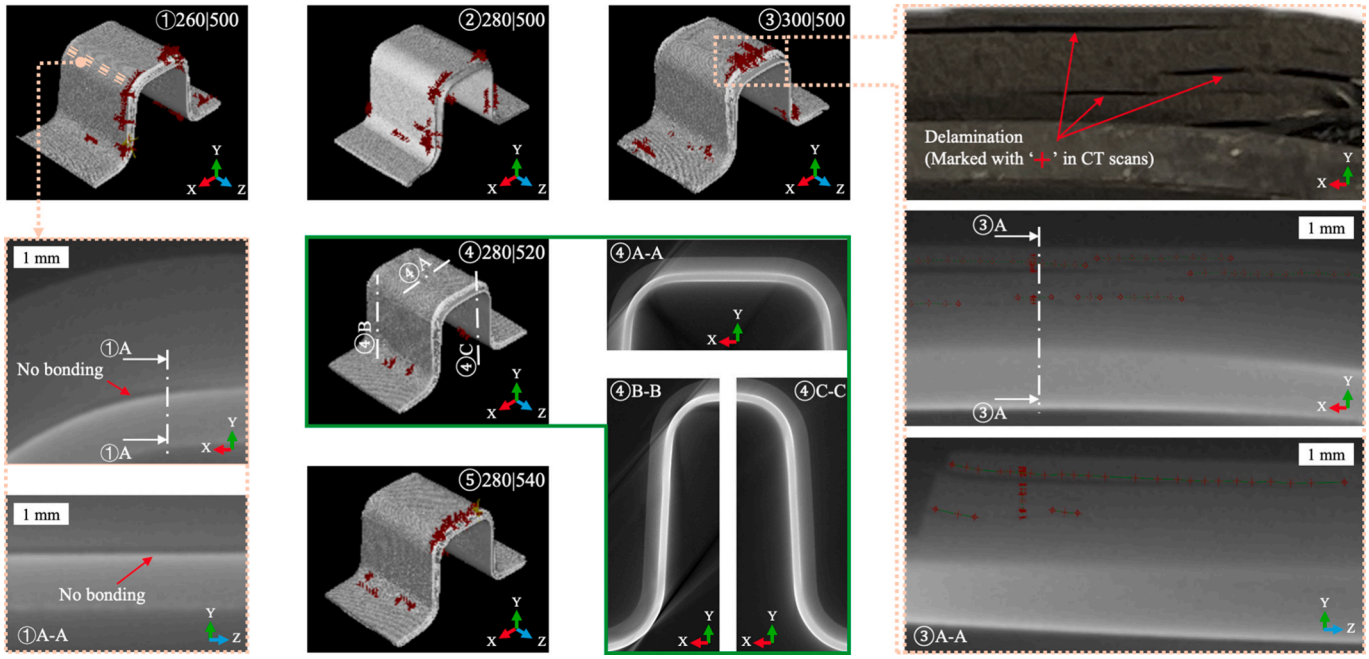


Fig. 22. Effects of stacking temperature on the formed FML components: bonding and delamination (CT scan resolution: 50 μm, target stacking temperatures indicated in the format of CF/PEEK|22MnB5, delamination identified with red markers '+' in cross-sections, as exampled with trial ③, and shown as scattered marks in the 3D reconstruction of each component).

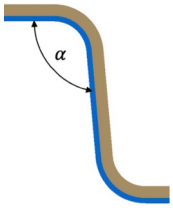
temperatures were relatively lower, but such defects were eliminated by further increasing stacking temperatures of either CF/PEEK or 22MnB5. Further increasing the CF/PEEK stacking temperature to 300 °C, or increasing the 22MnB5 stacking temperature to 540 °C, caused aggravated delamination in the top region in trials ③ and ⑤, respectively. As the CF/PEEK was formed at higher temperatures, fibres reorientated more easily, causing greater inter-ply sliding, while the movement of plies closer to the 22MnB5 was constrained due to friction. This resulted in a lateral strain mismatch, thereby causing more severe delamination, as well as the bias-extension-like necking, as shown in Fig. 21. Bottom radii delamination exists in all as-formed components due to the lack of normal pressure. However, these regions were considered less critical as they are usually trimmed off for the final component and can be addressed by introducing normal pressure with an improved tool design.

Overall, the CT results support the simulation-based process-window selection. The stacking temperatures of 280 °C for CF/PEEK and 520 °C for 22MnB5 produced the best post-form interfacial quality, combining sound metal-composite bonding with low inter-ply delamination.

Geometric distortion (springback and spring-in)

The stacking temperature also affects the geometric accuracy of the as-formed FML components, specifically the springback and spring-in effects. The geometric accuracy is evaluated by the side wall angle, measured from the CT scan results using myVGL 2023.4. Table 11 lists the measured angle, together with a schematic illustration. As the

Table 11
Side wall angles of the as-formed components.



22MnB5 (°C)	CF/PEEK (°C)		
	260	280	300
500	①96.20°	②93.43°	③92.83°
520		④92.96°	
540		⑤92.89°	

components are symmetric, only the angle from one side is measured.

The springback ratio (K) is calculated using Eq. (10):

$$K = \frac{\alpha_m - \alpha_d}{\alpha_d} \cdot 100\% \tag{10}$$

where the measured side-wall angle (α_m) is compared with the designed angle (α_d) of 95°. A positive value indicates springback, while a negative value indicates spring-in. Fig. 23 shows that increasing the stacking temperature of either CF/PEEK or 22MnB5 reduced the springback ratio and, at higher temperatures, introduced spring-in. Compared to the CF/PEEK stacking temperature, the 22MnB5 stacking temperature shows negligible influence on the geometrical accuracy. This is expected as hot-stamped 22MnB5 generally exhibits high dimensional accuracy [28]. Spring-in, on the other hand, has been a widely reported issue in thermoplastic-based composite processing [59]. The observed geometric distortion, especially the decreasing springback ratio as the CF/PEEK stacking temperature increases, is therefore mainly attributed to the thermomechanical and thermal response of the CF/PEEK laminate.

The through-thickness temperature gradient, as revealed by the process simulations, provides a valuable insight to qualitatively interpret the physics in the geometric distortion trend under different stacking temperature combinations. As simulations suggest that with stacking temperatures of 260 °C for CF/PEEK and 500 °C for 22MnB5, a relatively larger portion of the CF/PEEK is not melted when forming begins. This lower forming temperatures limits the inter-ply slip, a crucial mechanism for accommodating the strain difference between the outer and inner layers in forming multilayer sheets [60,61]. The resultant residual stress causes springback, as exhibited in trial ①. Increasing the CF/PEEK stacking temperature melts or softens more plies and increases the laminate forming temperature, but cooling from a higher temperature also increases thermal contraction and therefore promotes spring-in [62].

For semicrystalline thermoplastic composites such as CF/PEEK, cooling rate can also influence spring-in through crystallisation. Lower cooling rates can increase crystallinity [30,63], which increases matrix stiffness [30,64] and can increase spring-in. This mechanism may explain the larger spring-in observed in trials ②, ④ and ⑤ when the 22MnB5 stacking temperature was increased. However, because the steel-temperature increment was only 20 °C between adjacent conditions, its effect was smaller than that of the CF/PEEK stacking temperature. Tool compensation, commonly used to address springback in metal forming [65], could be applied to FML forming to improve the final shape. However, geometric compensation alone would not remove residual stresses, which may affect the metal-composite interface during service. A more robust route is to optimise the stacking temperatures and in-die cooling rate so that inter-ply slip, thermal contraction and crystallisation are better balanced. A post-form heat treatment can also help alleviate residual stresses primarily attributed to the FRP and the paint bake process in automotive manufacturing may be utilised.

Other factors, including lay-up anisotropy and component geometry [66–68], can also contribute to geometric distortion. Future modelling that incorporates thermal expansion, contraction and thermoplastic crystallisation would improve prediction of springback and spring-in and support process optimisation.

4.2.3. Hardness of the 22MnB5 in the as-formed components

The successful integration of the LT hot stamping in the one-shot FML forming process can also be demonstrated by examining the post-form hardness of the 22MnB5 layer. Rather than aiming to derive a continuous hardness trend across all stacking temperatures, the hardness of trial ④, which was formed at preferred target stacking temperatures of 280|520 °C, was measured to confirm the successful martensitic hardening. Fig. 24 shows that the hardness was measured across the profile of the demonstrator, with each position measured three times. The minimum and average hardness values were 437 HV10 and 453.3 HV10, respectively. This hardness level is indicative of an over 90 % martensitic transformation, corresponding to a UTS of approximately 1400 MPa [25]. Although a full-martensite microstructure was not achieved, due to the overall cooling rate being slightly inadequate attributed to the limitation in manual material transfer, the feasibility of one-shot forming technology for ultra-high strength FML components has been proven. Facilitating a more rapid transfer process by utilising advanced automated material handling systems with more precise process control is a promising solution for fully releasing the potential of this technology in further improving the post-form performance.

Taken together, the numerical and experimental results show that the preferred condition is not simply the highest available stacking temperatures. The process window requires sufficient heat from 22MnB5 to melt and bond the CF/PEEK surface, but not so much heat that the hotter laminate response promotes inter-ply delamination, localised necking or excessive spring-in. This balance explains why the

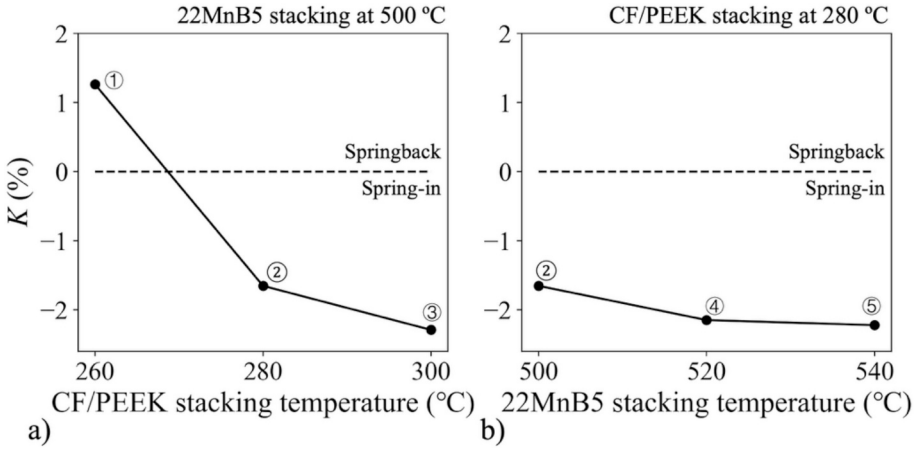


Fig. 23. Springback ratio (K) of trials with a) 22MnB5 target stacking temperature of 500 °C and b) CF/PEEK target stacking temperature of 280 °C.

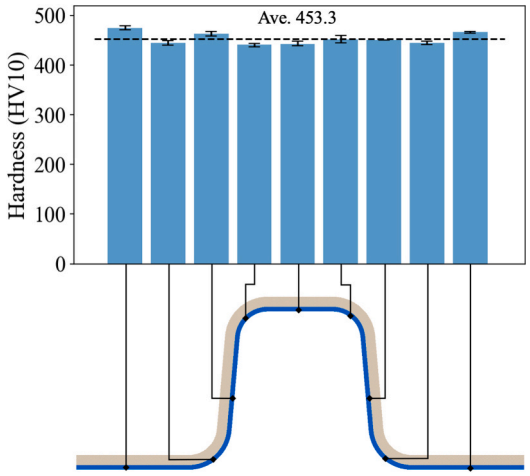


Fig. 24. Hardness distribution of the 22MnB5 layer in the as-formed component.

280|520 °C condition produced the best overall combination of bonding, delamination resistance, geometric quality and steel hardness within the tested conditions.

5. Conclusions

This study presents a proof of concept for a novel one-shot forming route for ultra-high-performance multi-material structures utilising CF/PEEK and 22MnB5. For the first time, an ultra-high-strength steel (UHSS) sheet is hot stamped, quenched and bonded to a fibre-reinforced thermoplastic laminate within a single shot forming cycle, producing fibre–metal laminate (FML) components without an adhesive layer. The process integrates composite and metal forming, metal–composite bonding and martensitic hardening of the steel within a single-step stamp-forming process. Its cycle time is at the seconds level, comparable to the productivity of industrial metal hot stamping, indicating potential scalability and ease of adoption by existing metal hot-stamping lines. By enabling solid-state forming of the composite in cold dies, this technology also presents an opportunity to reduce energy demand compared with conventional composite forming routes. The main conclusions are as follows:

1) A separate-heating, pre-cooling and one-shot forming route was formulated to reconcile the process requirements of thermoplastic composite solid-state forming and UHSS hot stamping. The route locally melts the CF/PEEK surface at the metal-composite interface while keeping the remaining laminate within a solid-state forming window and maintaining the 22MnB5 thermal path required for martensitic hardening.
2) A two-step stacking-forming simulation was developed as a process-window design tool. The stacking step predicted the through-thickness temperature evolution, while the forming step ranked the inter-ply delamination risk using cohesive surface damage. The model identified 280 °C for CF/PEEK and 520 °C for 22MnB5 as the preferred stacking temperatures within the investigated window.
3) Forming trials and post-form assessment supported the predicted preferred window. At 520 °C for 22MnB5 and 280 °C for CF/PEEK, the demonstrator showed good metal–composite interfacial quality, low CT-detected inter-ply delamination, a formed geometry close to the target U-profile and a 22MnB5 hardness of 453.3 HV10, corresponding to an estimated ultimate tensile strength above 1400 MPa. Lower temperatures led to insufficient interfacial bonding, whereas higher temperatures promoted localised necking and additional delamination in the composite layer. Residual springback/spring-in indicated scope for further optimisation of the thermal history, transfer control and forming conditions.

Overall, this work establishes a new manufacturing route for UHSS-based FML components by integrating composite and metal forming, metal–composite bonding and steel quench hardening within a single shot forming. The significance of the approach lies in using a controlled transient thermal gradient to reconcile the otherwise incompatible processing requirements of UHSS hot stamping and thermoplastic composite forming. By eliminating separate forming and post-form adhesive bonding, the route provides a promising basis for faster, more integrated manufacturing of high-performance lightweight multi-material structures for automotive applications. The present proof of concept demonstrates the feasibility of extending hot-stamping-based manufacturing towards adhesive-free thermoplastic FML components for automotive applications. Future work can be focused on quantitative assessment of the interfacial bonding quality, structural validation, simulation model validation, and further process optimisation for geometry-specific industrial conditions.

CRediT authorship contribution statement

Zerong Ding: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hongyan Wang:** Writing – review & editing, Methodology, Formal analysis. **Chunyi Gao:** Writing – review & editing, Methodology, Formal analysis. **Nan Li:** Writing – review & editing, Project administration, Supervision, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During preparation of this revised manuscript, the authors used an AI writing assistant to support language editing. The authors reviewed and edited all AI-assisted text and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zerong Ding reports financial support was provided by Shougang-Imperial Lab for Lightweight Steel-based System for Impact Resistant Automotive Applications. Hongyan Wang reports financial support was provided by Chinese Scholarship Council (CSC). Chunyi Gao reports financial support was provided by Imperial College London.

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Data availability

Data will be made available on request.

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